

LOSO Experiment B_ypr_accel_lstm

LSTM only · YPR + Accel (both hands), no Nothing

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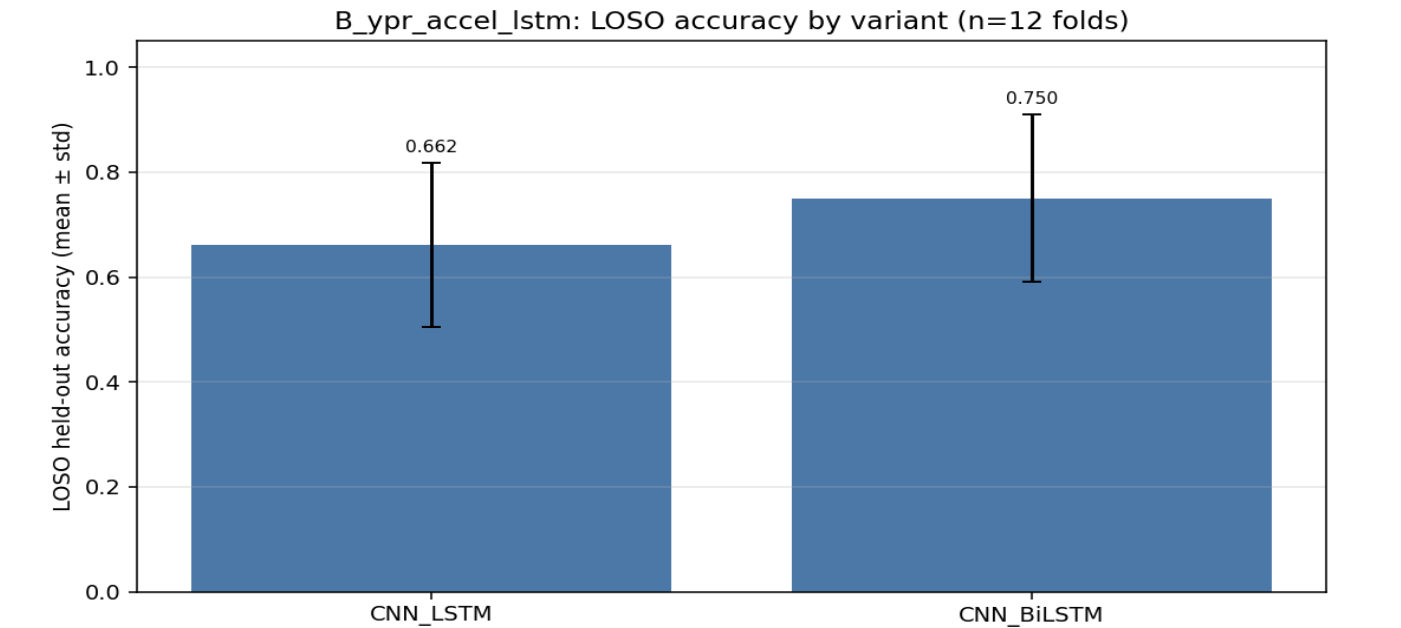
Experiment configuration

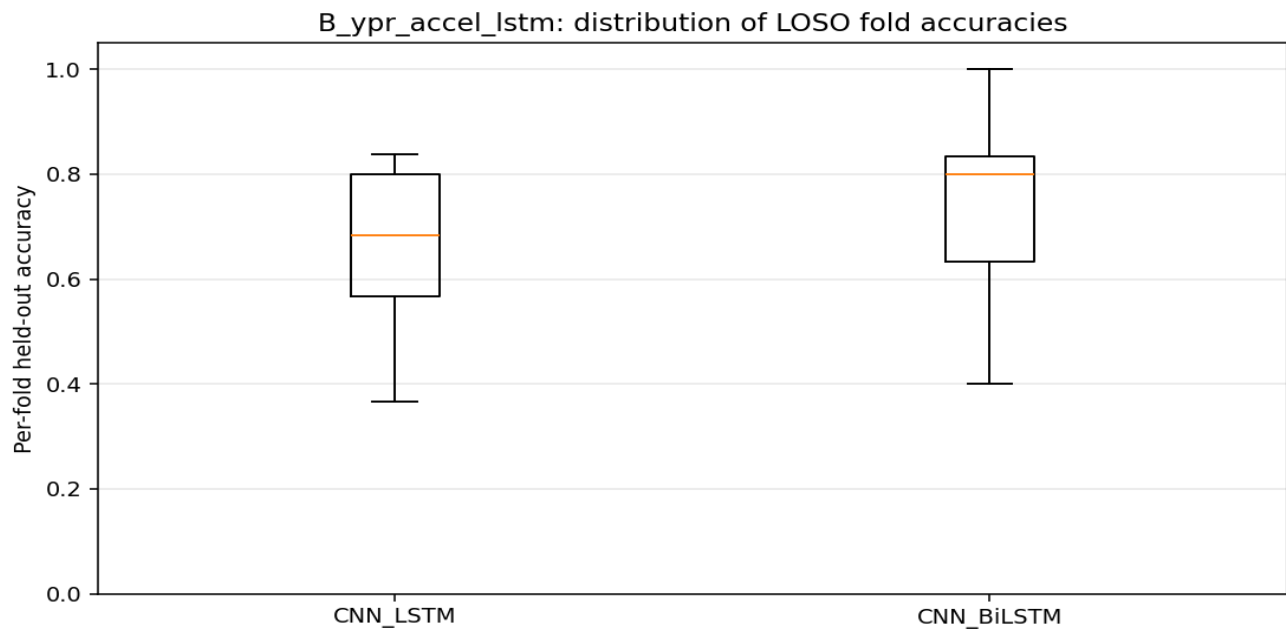
Class set	6 classes excluding Double_Nothing
Classes	Double_Lower, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor modalities	YPR, accel
Hands	left, right
Sensor channels	156
Sequence length	90 steps (resampled)
Filter	Butterworth 10.0Hz order 4
Normalisation	minmax
Augmentation	amplitude_scale + baseline_offset, 3 copies
Inner CV folds	3
Epochs · batch size	8 · 16
LOSO subjects	12: 11_Mansh, 12_Marcus, 14_StephenV2, 15_Tash, 16_Bella, 17_Raquel, 18_Bridgette, 1_Alan, 2_Alex, 3_Anghad, 4_Daniel, 6_Harry

LOSO summary across all subjects

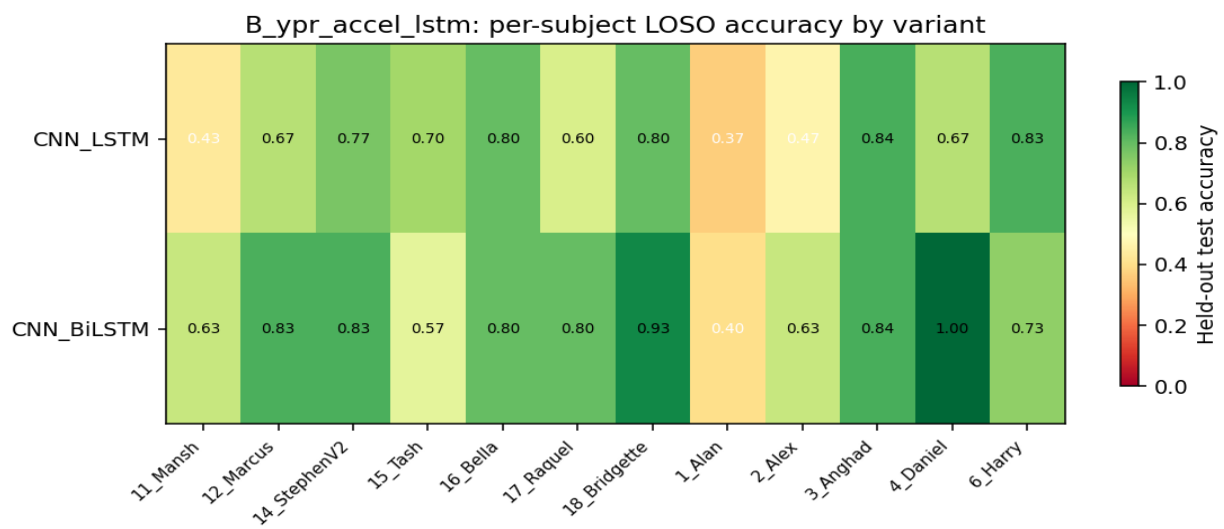
Variant	Params	LOSO mean	LOSO std	Min fold	Max fold	Inner CV mean (avg across folds)
CNN_LSTM	65,830	0.6616	0.1562	0.3667	0.8387	0.7701
CNN_BiLSTM	102,950	0.7504	0.1598	0.4000	1.0000	0.8265

Best variant by LOSO mean: **CNN_BiLSTM**





Per-subject LOSO accuracy heatmap



Per-fold detail (held-out subject = each row)

Held-out subj	Train trials	CNN_LSTM	CNN_BiLSTM
11_Mansh	331	0.433	0.633
12_Marcus	331	0.667	0.833
14_StephenV2	331	0.767	0.833
15_Tash	331	0.700	0.567
16_Bella	331	0.800	0.800
17_Raquel	331	0.600	0.800
18_Bridgette	331	0.800	0.933
1_Alan	331	0.367	0.400
2_Alex	331	0.467	0.633
3_Anghad	330	0.839	0.839
4_Daniel	331	0.667	1.000
6_Harry	331	0.833	0.733

Variant: CNN_LSTM

Params: 65,830

LOSO mean accuracy: 0.6616 ± 0.1562 (min 0.3667, max 0.8387)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	5/5	3/5	5/5	0/5	0/5	0/5	0.433
12_Marcus	5/5	5/5	5/5	0/5	5/5	0/5	0.667
14_StephenV2	5/5	5/5	5/5	3/5	5/5	0/5	0.767
15_Tash	5/5	5/5	5/5	5/5	0/5	1/5	0.700
16_Bella	5/5	5/5	0/5	4/5	5/5	5/5	0.800
17_Raquel	5/5	5/5	5/5	0/5	3/5	0/5	0.600
18_Bridgette	5/5	5/5	5/5	3/5	5/5	1/5	0.800
1_Alan	5/5	0/5	0/5	1/5	5/5	0/5	0.367
2_Alex	3/5	4/5	5/5	0/5	2/5	0/5	0.467
3_Anghad	3/5	5/5	5/5	2/5	6/6	5/5	0.839
4_Daniel	5/5	5/5	5/5	0/5	4/5	1/5	0.667
6_Harry	5/5	4/5	5/5	2/5	5/5	4/5	0.833
MEAN	0.933	0.850	0.833	0.333	0.733	0.283	0.662

Variant: CNN_BiLSTM

Params: 102,950

LOSO mean accuracy: 0.7504 ± 0.1598 (min 0.4000, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	5/5	4/5	5/5	5/5	0/5	0/5	0.633
12_Marcus	5/5	5/5	5/5	1/5	5/5	4/5	0.833
14_StephenV2	0/5	5/5	5/5	5/5	5/5	5/5	0.833
15_Tash	0/5	5/5	2/5	5/5	5/5	0/5	0.567
16_Bella	5/5	5/5	2/5	5/5	5/5	5/5	0.800
17_Raquel	5/5	5/5	5/5	4/5	5/5	0/5	0.800
18_Bridgette	3/5	5/5	5/5	5/5	5/5	5/5	0.933
1_Alan	5/5	0/5	0/5	2/5	5/5	0/5	0.400
2_Alex	5/5	4/5	5/5	0/5	5/5	0/5	0.633
3_Anghad	0/5	5/5	5/5	5/5	6/6	5/5	0.839
4_Daniel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
6_Harry	5/5	5/5	5/5	0/5	2/5	5/5	0.733
MEAN	0.717	0.883	0.817	0.650	0.867	0.567	0.750